

○ Affirmative Steps

At the same time, policymakers should take affirmative steps to facilitate open spectrum. Most current unlicensed wireless services, including WiFi, operate in the 2.4 GHz and 5 GHz unlicensed bands. These bands are relatively narrow, at high frequencies that limit their propagation, and subject to many established competing uses. Though unlicensed devices can coexist in seemingly crowded spectrum, their ability to do so is not absolute. Moreover, WiFi's software protocols don't have the adaptive and cooperative characteristics of truly scalable unlicensed networks. Current FCC rules have done a reasonable job of setting conditions that allow for innovation and market growth, but more is needed.

The US government should follow a four-step program to make open spectrum a reality:

- Develop rules to foster more effective cooperation among unlicensed users
- Set aside more spectrum for unlicensed uses
- Eliminate restrictions on non-intrusive underlay techniques across licensed bands
- Promote experimentation and research in unlicensed wireless technology

All of the elements are important. WiFi, other unlicensed technologies in designated bands, and underlay are all part of the answer. Furthermore, the mix will change over time. Existing unlicensed bands are delivering value today. However, newer approaches designed from the ground up for open spectrum will be the long-term winners.

The only way to allow market forces to determine the best solutions is to give alternative approaches a chance. By announcing its intention to move forward with a comprehensive open